

Data

12 frozen encoders

Stage 1 analyses

mRNA stability

CodonBERT benchmark, n = 981

SAME input: one coding sequence read
as DNA, as RNA, and translated

target: stability compare: BioLangFusion

GTEx expression

IsoFormer pilot, n = 999 + 996

DISTINCT inputs: 6 kb DNA window,
transcript, protein

target: 30 tissues compare: IsoFormer

DNA

NT 500M

1 token = 6-mer
1280-d embedding

NT v2 100M

1 token = 6-mer
512-d embedding

DNABERT-2

1 token = byte-pair
768-d embedding

HyenaDNA

1 token = nucleotide
256-d embedding

RNA

RNA-FM

1 token = nucleotide
640-d embedding

RiNALMo

1 token = nucleotide
640-d embedding

mRNA-FM

1 token = codon
1280-d embedding

CaLM

1 token = codon
768-d embedding

protein

ESM-2 8M

1 token = amino acid
320-d embedding

ESM-2 35M

1 token = amino acid
480-d embedding

ESM-2 150M

1 token = amino acid
640-d embedding

ProtBERT

1 token = amino acid
1024-d embedding

Track A: Decodability

ridge probes for stability, expression,
GC / GC3, structure; concatenation gains

Track B: Geometry

CKA, mutual k-NN, CCA retrieval, RSA,
before and after the composition control

Track C: Diagnostics

layer-wise CKA, attention against motifs,
per-sequence alignment vs error

frozen · final layer · mean-pooled to one vector per sequence

evidence

Stage 2 · define alignment

Aligned = agreement on held-out sequences
AFTER the composition control. Reported per
pair, kept separate from complementarity.

Stage 3 · fusion architecture

Token-level fusion. Coupled Mamba on the
codon grid where biology supplies one; cross-
attention on GTEx. No alignment loss by default.

Stage 4 · benchmarking + method test

Must beat length, composition, concatenation
and the best single encoder. Then: do Stage 1
measures predict which pairs gain?